



Deep Learning & Computer Vision:

Lab session 1/2

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Synopsis

- In the two lab sessions to come you will develop 4 exercises, from very simple to a more advanced one developping a CNN classifying skin moles into seven cancer categories.
- You will code in python and use pytorch or keras tensorflow [1].
- You will a GPU-accelerated plateform running in cloud provided along with a Gmail account. Your code and data will be stored at your GDrive.

  Note : *You will be able to run your code from anywhere provided you have an internet connection and a navigator.*

1. according to your liking

Lab assignments

- The exercise assignments are provided as incomplete jupyter notebooks .ipynb containing all necessary helper functions and a pdf containing an example solution you should obtain. You only need to concentrate on coding the deep learning part.
- In lab sessions 5 and 6 you will develop 4 exercises

Lab session 5 :

1. Exercise 1 : a simple non-linear classifier using a few neurons.
2. Exercise 2 : a hand-written digits classifier with a dens net
3. Exercise 3 : a hand-written digits classifier with a CNN

A single-file pdf report is to be uploaded to moodle.

Lab session 6 :

4. Exercise 4 : a skin cancer classifier using a simple CNN

A pdf report is to be uploaded to moodle.

Make sure both names appear in the report for each team.

1. You need a Gmail account. Create one now if you do not have. Connect to your Gmail in a navigator. Then you can access the Colab platform at https://colab.research.google.com/ → (Tab Google Drive)
2. click to open the link `https://colab.research.google.com/notebooks/basic_features_overview.ipynb` (or type in the search prompt if not working)
3. From the colab, 'File' -> 'Save a Copy to Drive' of the `basic_feature_overview.ipynb` in your Drive. This saves a private copy of the file in a new directory "Colab Notebooks" in your drive. All your notebooks should reside in this special 'Colab Notebooks' directory in your GDrive to open and execute correctly.
4. Go through this file to discover all the Colab features.

1. This file is a python notebook `.ipynb`. Notebooks are *active* files allowing to mix images, text and code. You can even use Latex for handsome maths rendering. When the code is executed the results can either be saved or displayed in the notebook.
2. Have a look on the file. Different portions of the document are structured in *cells*. You can modify a text cell by double clicking on it. You can run a code cell by clicking inside the brackets `[ ]` on top left of the cell. Once a cell has been executed it is assigned a number in the brackets. The cells should be executed in the order of appearance.
3. The output appears below each cell (or an error flag  and a message below and if there was an error.
4. Various other commands related to the code execution are available in the menu.
5. Go on to execute one by one the cells of the notebook. Do not "execute all" because your interaction is needed from time to time.

Lab assignments (cntd.)

- Download from `Lab Session 5 (resources)` on your moodle the exercise assignments `.ipynb` and proceed with.
- The last cell in the lab assignment notebooks contain a code allowing to generate a pdf that can serve as a report.
- However, a single file pdf per lab session is to be uploaded to moodle. Make sure to concatenate all the exercise files into a single file.
- Extra time is allowed to finish. **(The deadline date is indicated on moodle.)**
- **The report is evaluated. Make sure you comment abundantly the report and the results. Answer all questions. Make sure both teammate names appear in the report for each team**.